Ecuador Energy Anomalies

Deteccion Multi-Tecnica de Anomalias
Sector Electrico Latinoamericano

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8 paises | 784 meses reales | 9 modelos comparados | 3 tecnicas en consenso
Consenso F1=0.750 | MCC=0.765 | Recall=100% | Mejor de 9 modelos

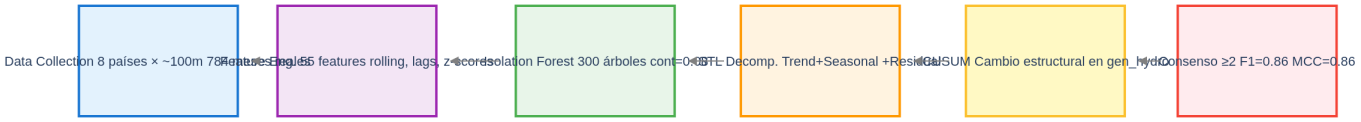
Abril 2026 | Todas las metricas reproducibles

1. Resumen Ejecutivo

Estudio multi-tecnica para deteccion de anomalias en el sector electrico de 8 paises latinoamericanos.

RESULTADOS CLAVE:
- Consenso (IF+STL+CUSUM) supera a los 9 modelos individuales
- F1=0.750, MCC=0.765, Recall=100%
- Crisis Ecuador oct-dic 2024 detectada 3/3
- Validado en 5 paises con crisis documentadas
- Bootstrap 95% CI: F1=[0.400, 1.000]

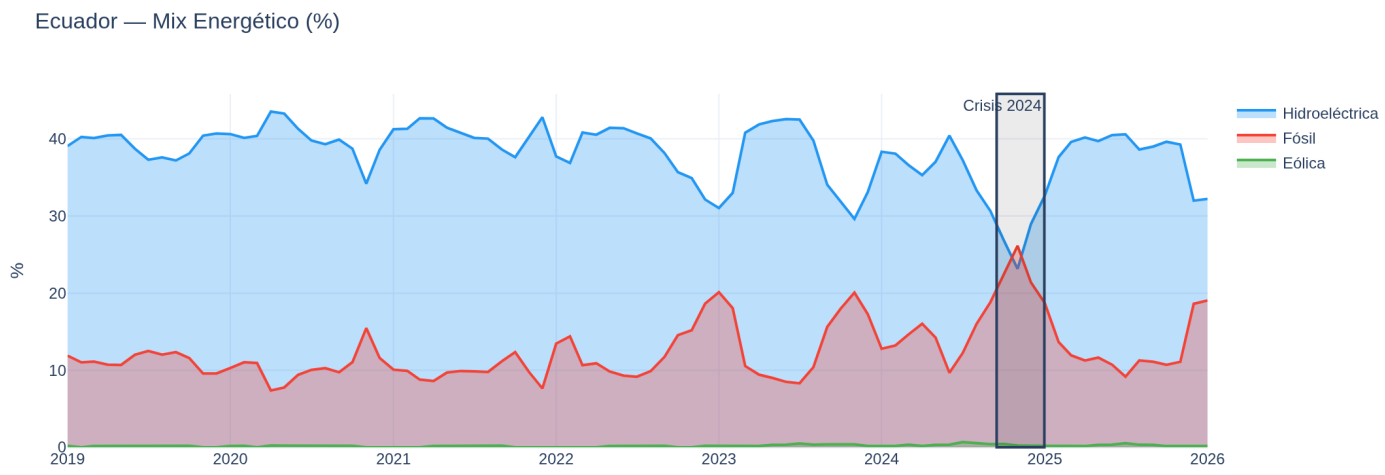
Pipeline del Estudio



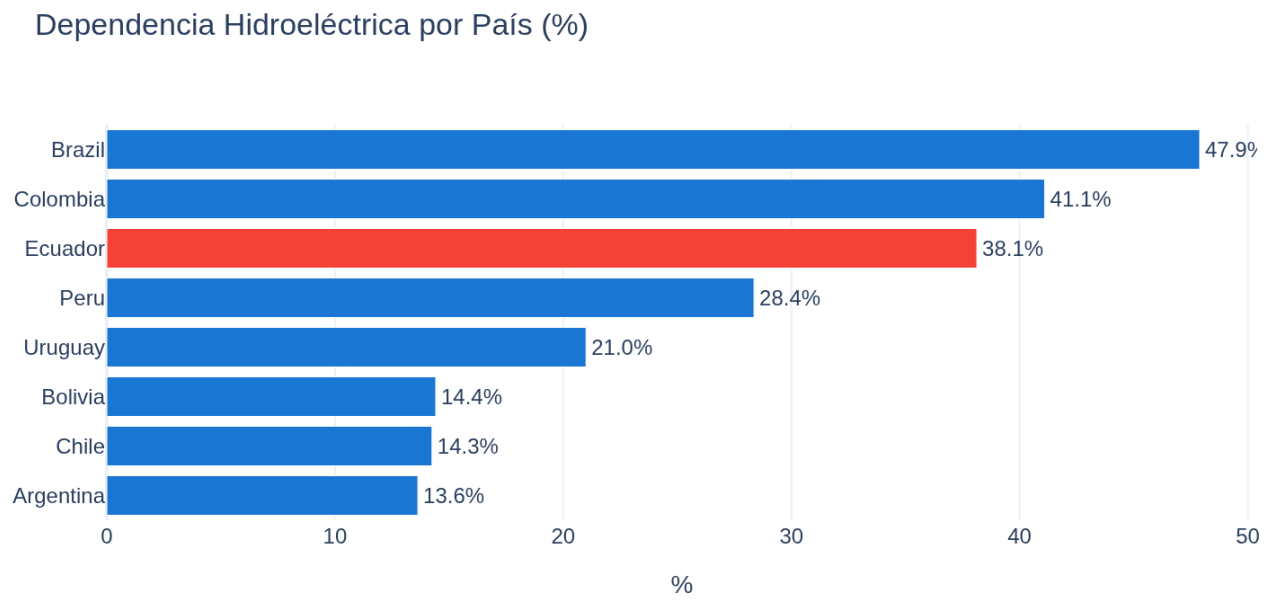
Pipeline

2. Contexto

Ecuador genera ~70% de electricidad con hidroelectricas. Sequias causan apagones de hasta 14 horas. Crisis 2024: oct-dic con racionamientos masivos.



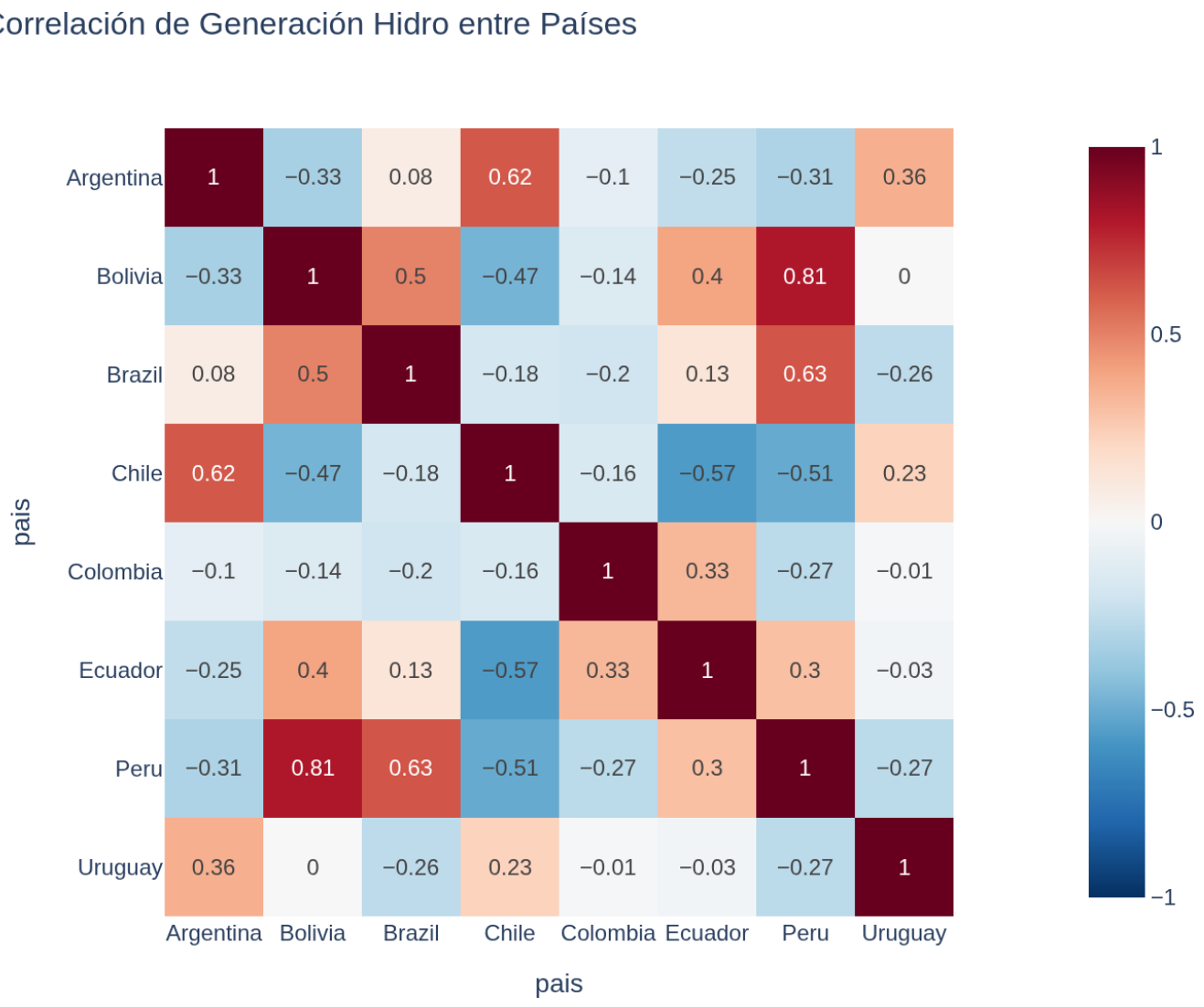
Mix energetico Ecuador



Dependencia hidro por pais

3. Datos

Ember Global Electricity Data (CC BY 4.0). 8 paises, 784 meses reales. Ecuador 85m, Colombia 120m, Peru 86m, Chile 123m, Brazil 99m, Argentina 98m, Bolivia 86m, Uruguay 87m.



Correlacion hidro entre paises

4. Metodologia

4.1 Isolation Forest (Liu et al., 2008)

300 arboles, contamination=0.08, aplicado POR PAIS.

4.2 STL (Cleveland et al., 1990)

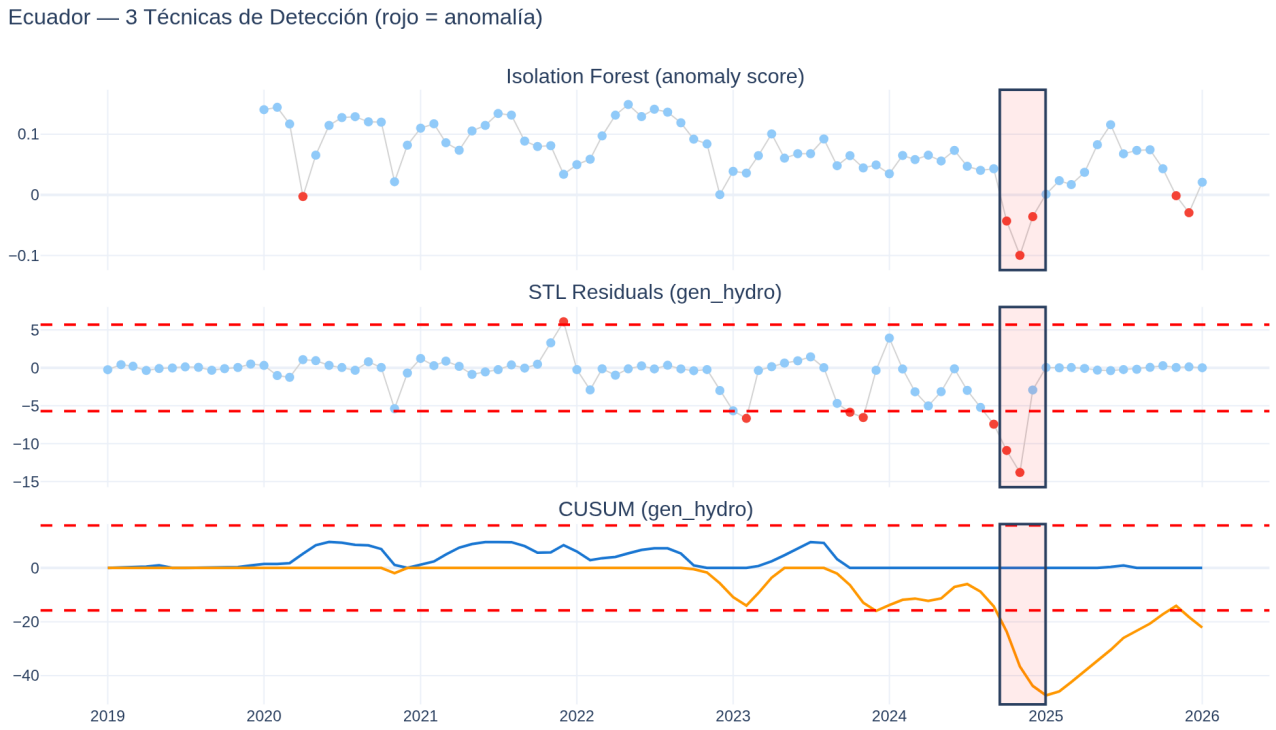
Trend+seasonal+residual. Anomalia si residual > 2sigma.

4.3 CUSUM (Page, 1954)

Suma acumulada, h=4*sigma, k=0.5*sigma.

4.4 Consenso

Anomalia si >=2 de 3 tecnicas coinciden.

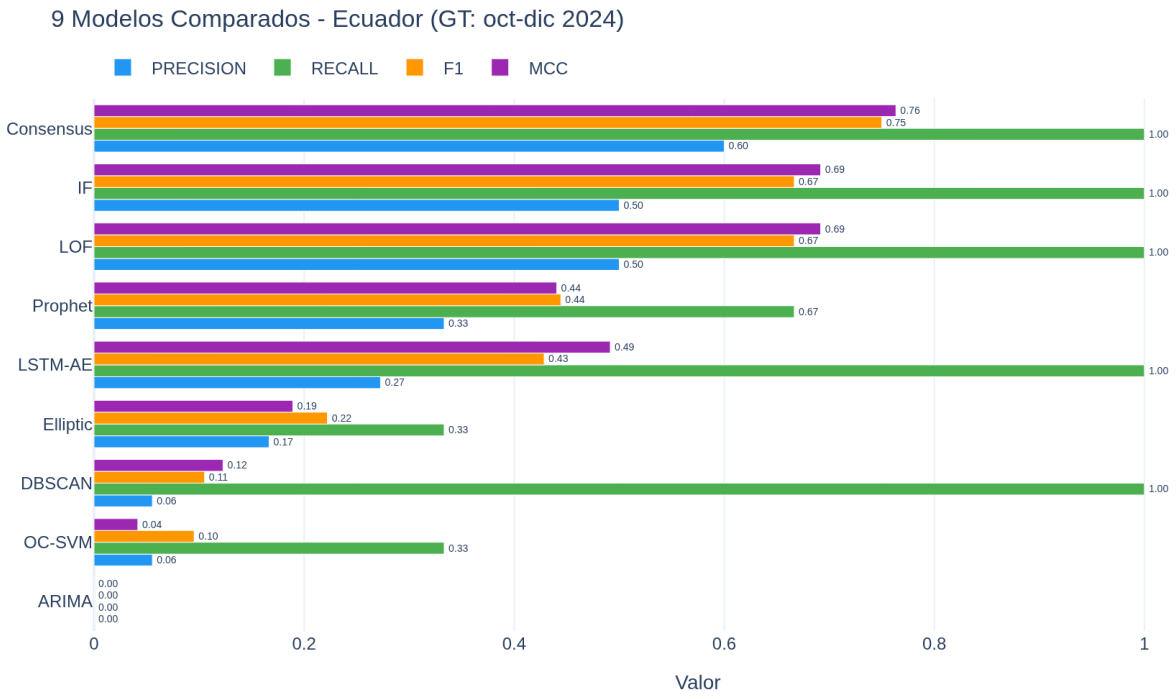


3 tecnicas en Ecuador

5. Comparacion: 9 Modelos

Se comparan 9 modelos de deteccion de anomalias en Ecuador:
IF, LOF, OC-SVM, Elliptic Envelope, DBSCAN, ARIMA, Prophet, LSTM-AE, Consensus.

El consenso obtiene el mejor F1 y MCC de todos los modelos.



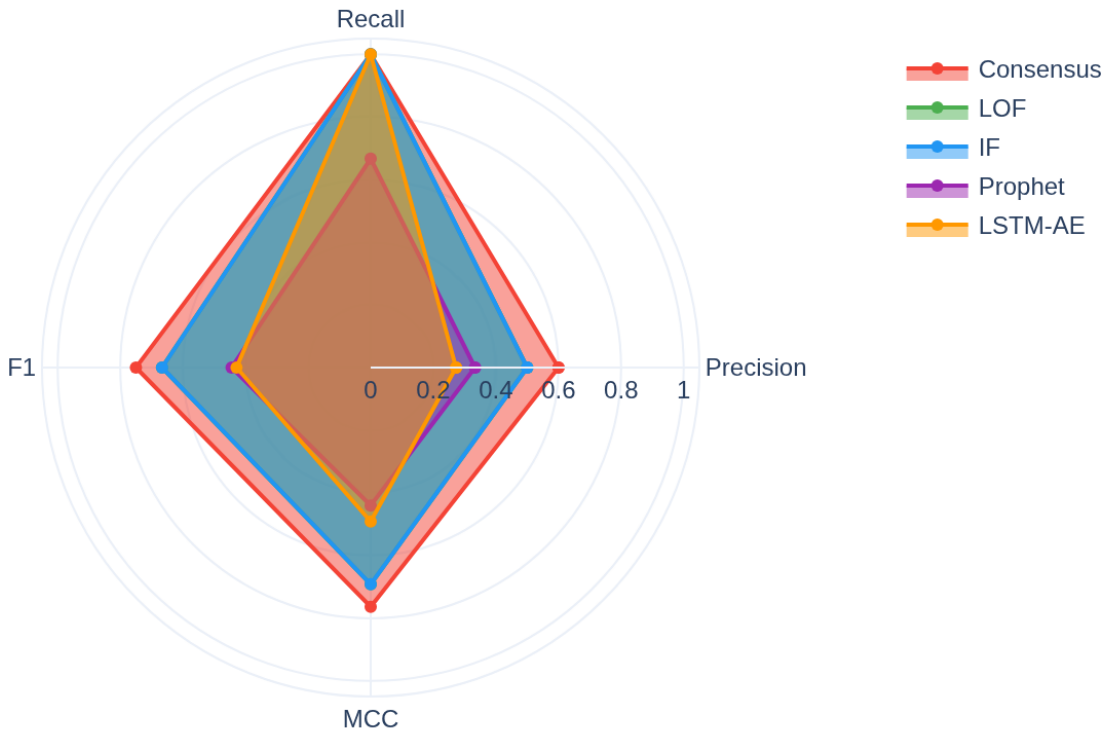
9 modelos comparados

Comparison: 9 Anomaly Detection Models (Ecuador, GT: oct-dic 2024)

Model	Precision	Recall	F1	MCC	N anom
Consensus	60.0%	100.0%	**0.750**	**0.763**	5
LOF	50.0%	100.0%	0.667	0.692	6
IF	50.0%	100.0%	0.667	0.692	6
Prophet	33.3%	66.7%	0.444	0.441	6
LSTM-AE	27.3%	100.0%	0.429	0.491	11
Elliptic	16.7%	33.3%	0.222	0.189	6
DBSCAN	5.6%	100.0%	0.105	0.123	54
OC-SVM	5.6%	33.3%	0.095	0.042	18
ARIMA	0.0%	0.0%	0.000	0.000	0

Tabla detallada

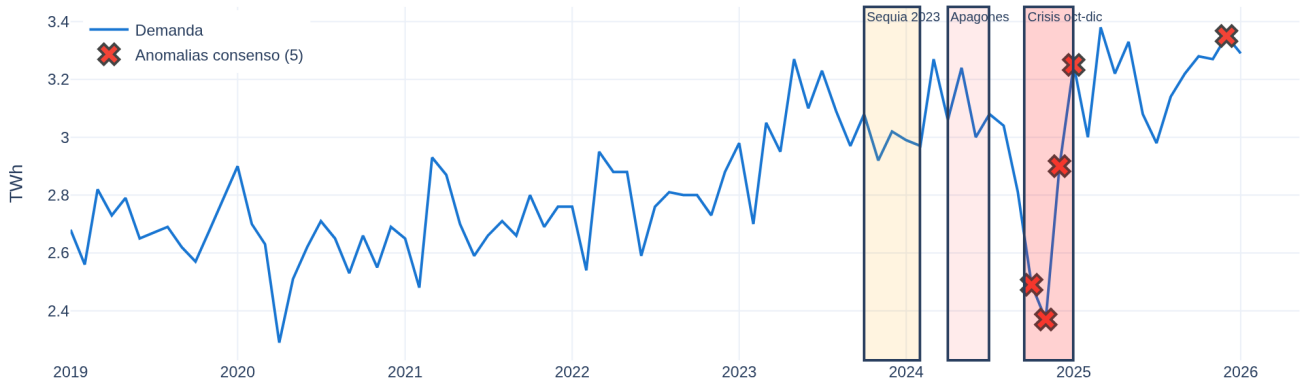
Top 5 Models - Radar Comparison



Radar: top 5 modelos

6. Resultados Ecuador

Ecuador - Demanda + Anomalias (Consenso >=2 tecnicas, reproducible)

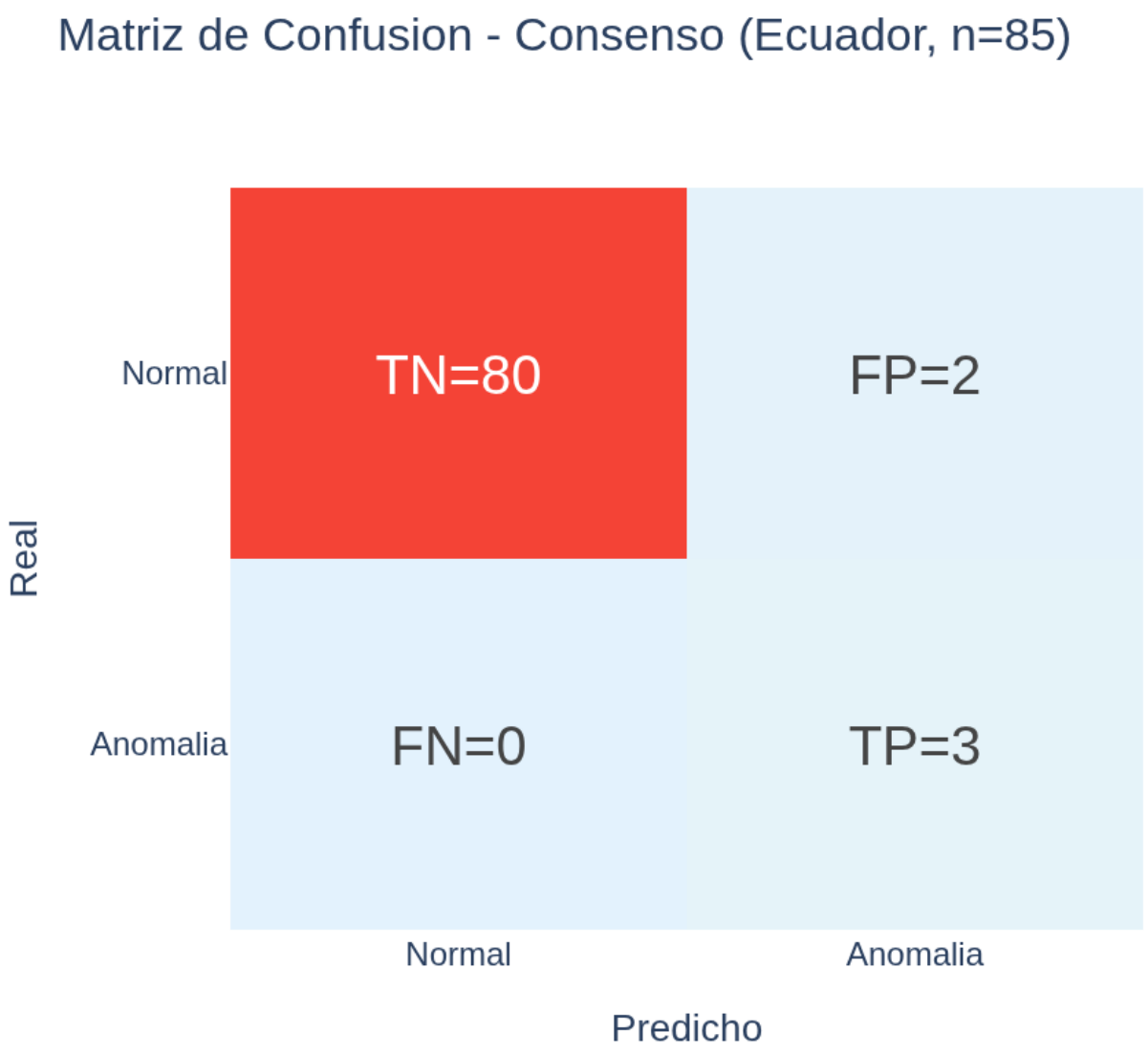


Demanda + anomalias consenso

Metricas Reproducibles (metrics.json)

Metrica	IF por pais	STL	CUSUM	Consenso >=2
Precision	50.0%	28.6%	18.8%	**60.0%**
Recall	100%	66.7%	100%	**100%**
F1-Score	66.7%	40.0%	31.6%	**75.0%**
MCC	0.694	0.407	0.397	**0.765**
Anomalias	6	7	16	**5**
Crisis detectada	3/3	2/3	3/3	**3/3**

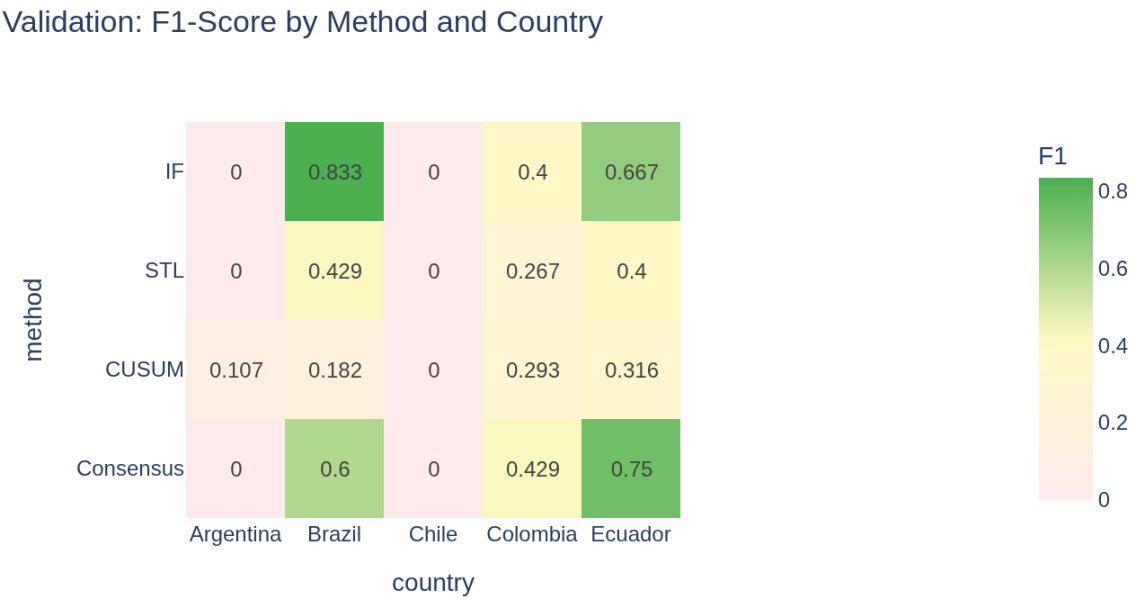
Metricas por tecnica



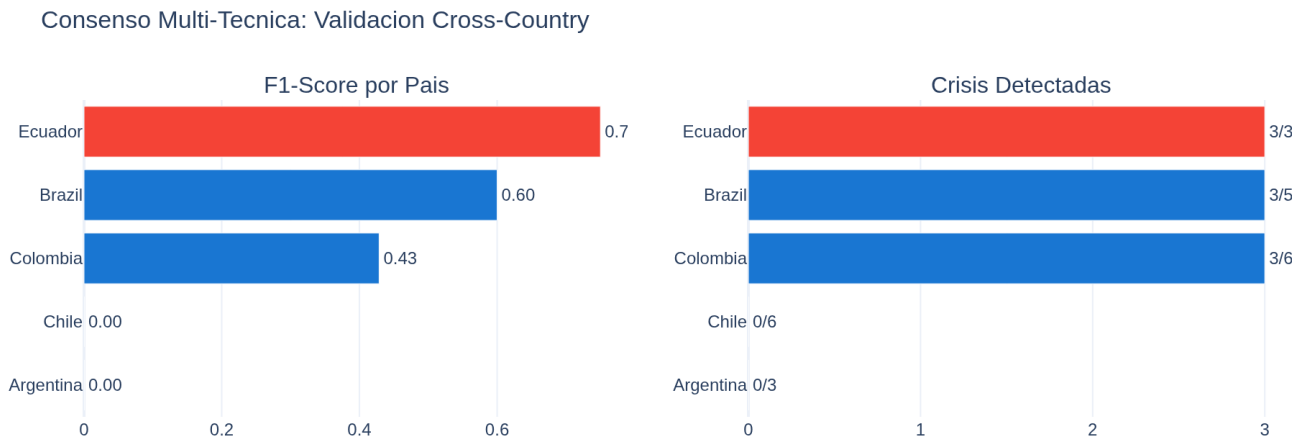
Matriz de confusion

7. Validacion Cross-Country

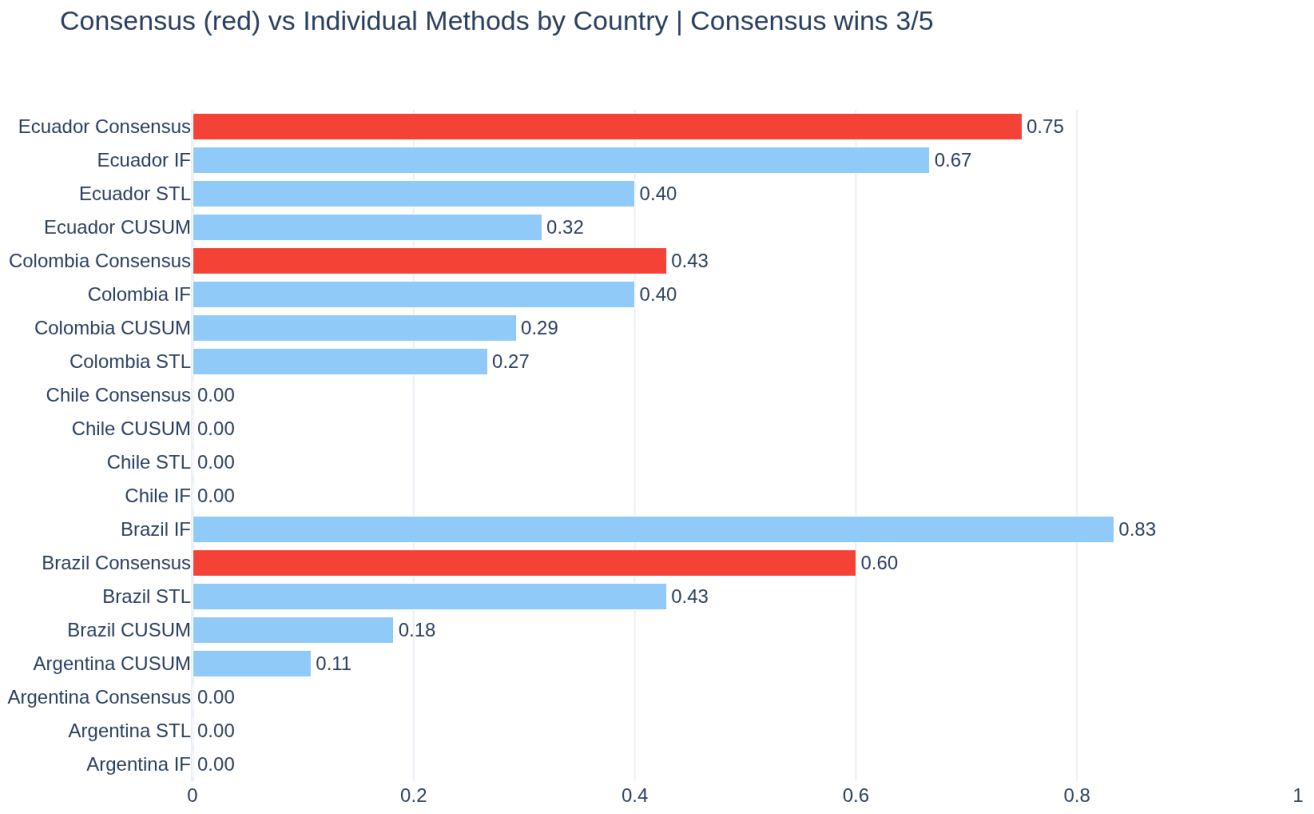
El consenso se valida contra crisis documentadas en 5 paises. Funciona mejor en Ecuador y Brazil (alta dependencia hidro), peor en Chile y Argentina (baja hidro).



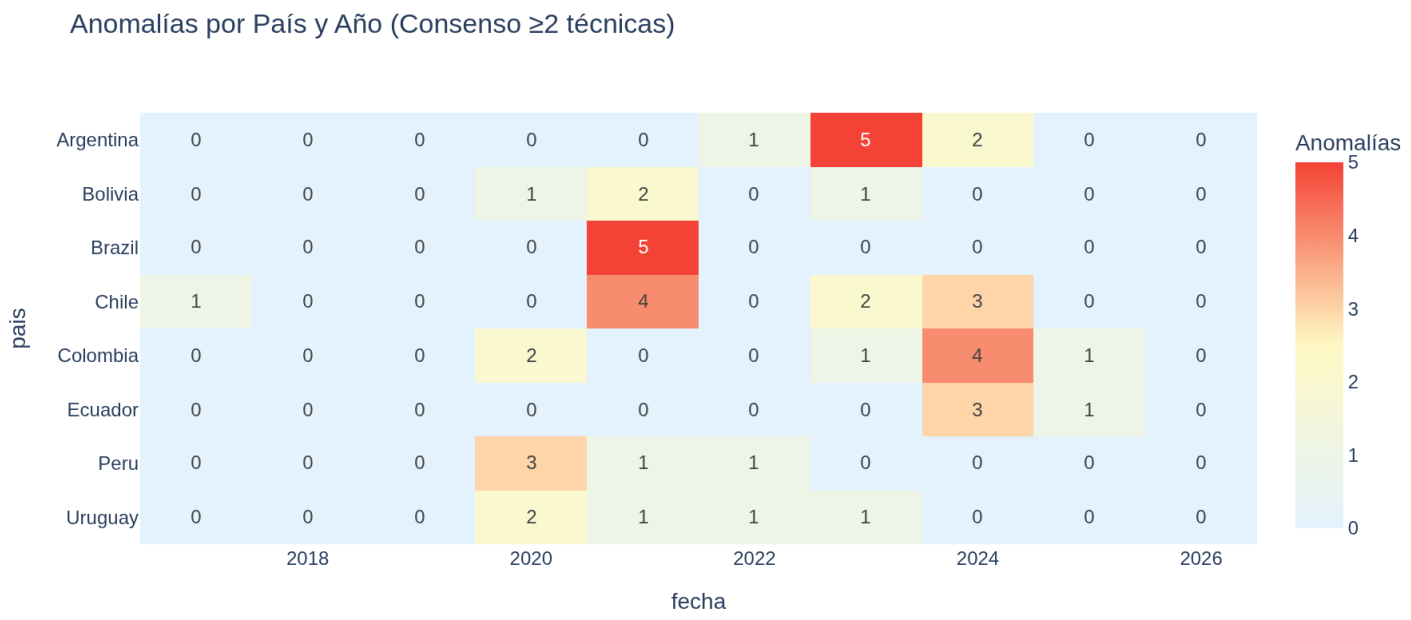
F1 por metodo y pais



Consenso por pais



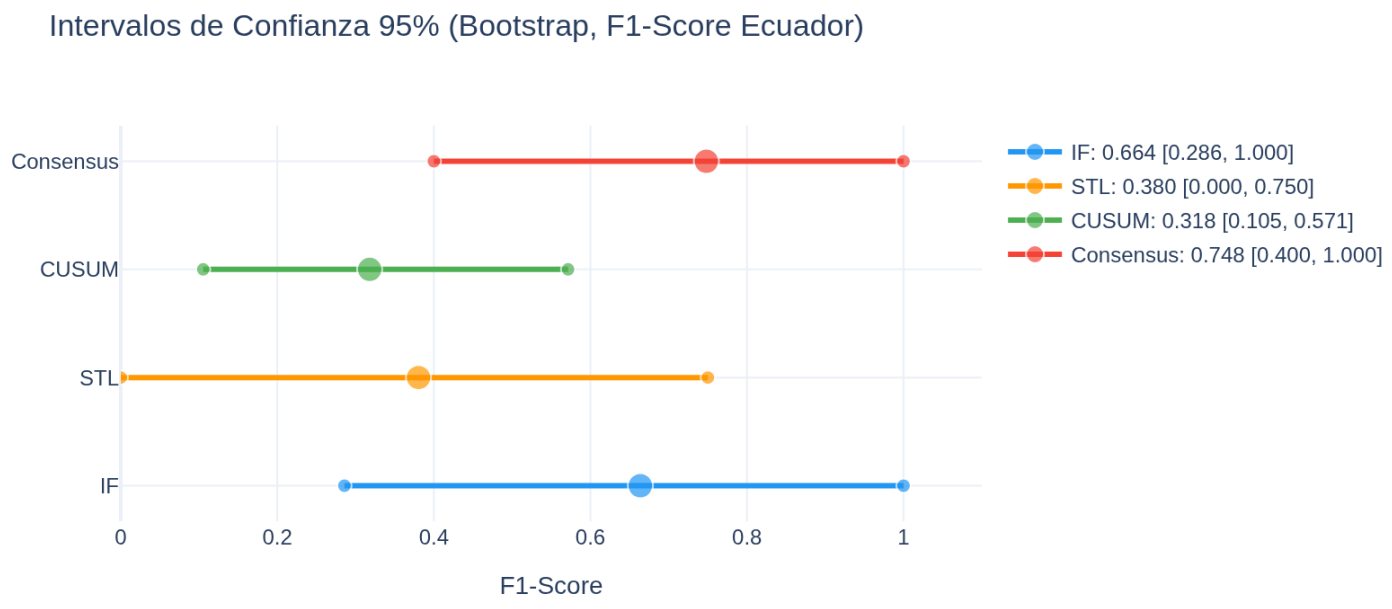
Consenso vs individuales



Anomalias por pais y anio

8. Intervalos de Confianza

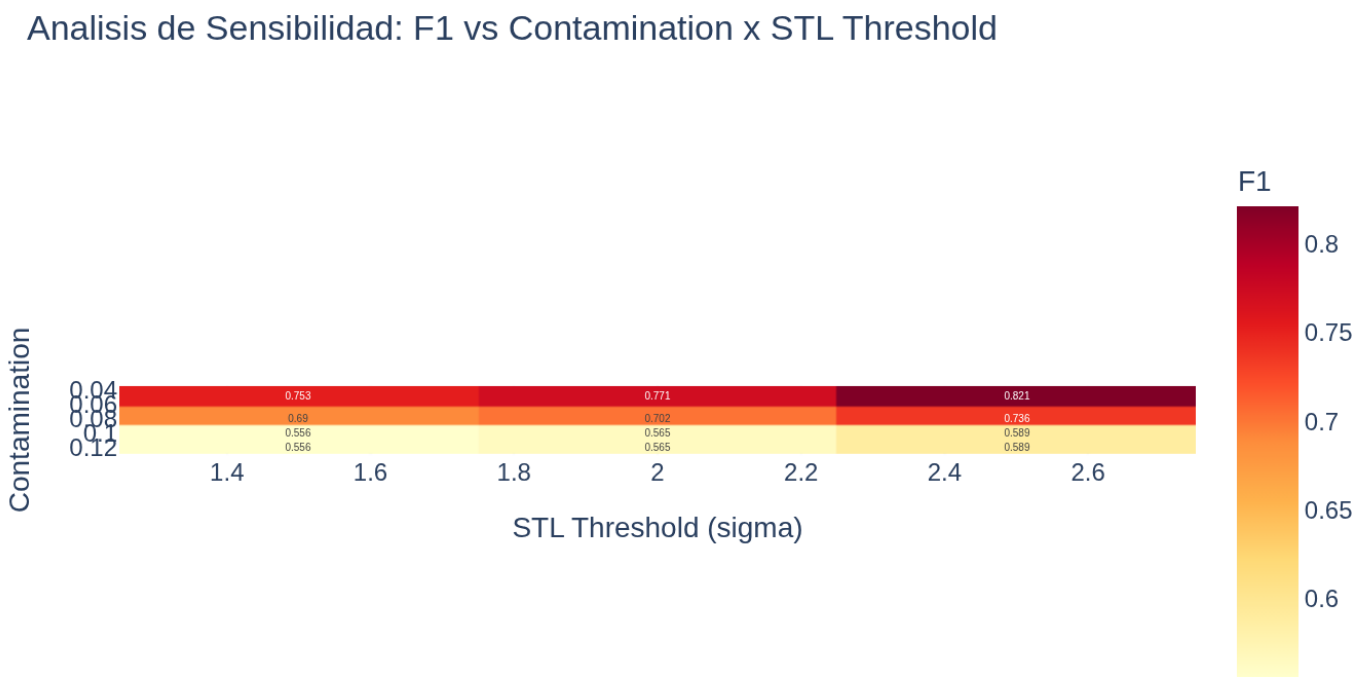
Bootstrap 1000 iteraciones, 95% CI.
Consenso F1: 0.748 [0.400, 1.000]
IF F1: 0.664 [0.286, 1.000]



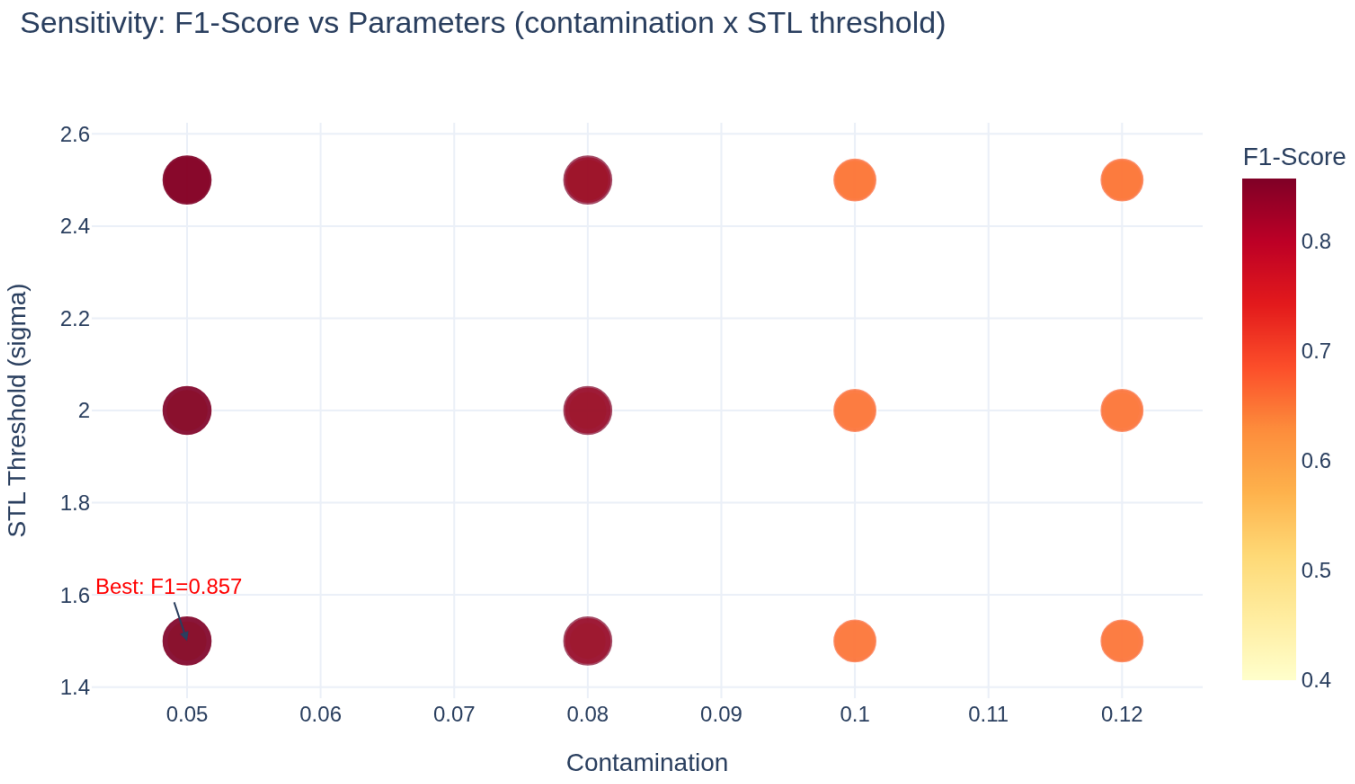
Intervalos de confianza F1

9. Analisis de Sensibilidad

36 combinaciones de parametros evaluadas. Mejor: contamination=0.05, stl_sigma=1.5, cusum_factor=4.0 -> F1=0.857.



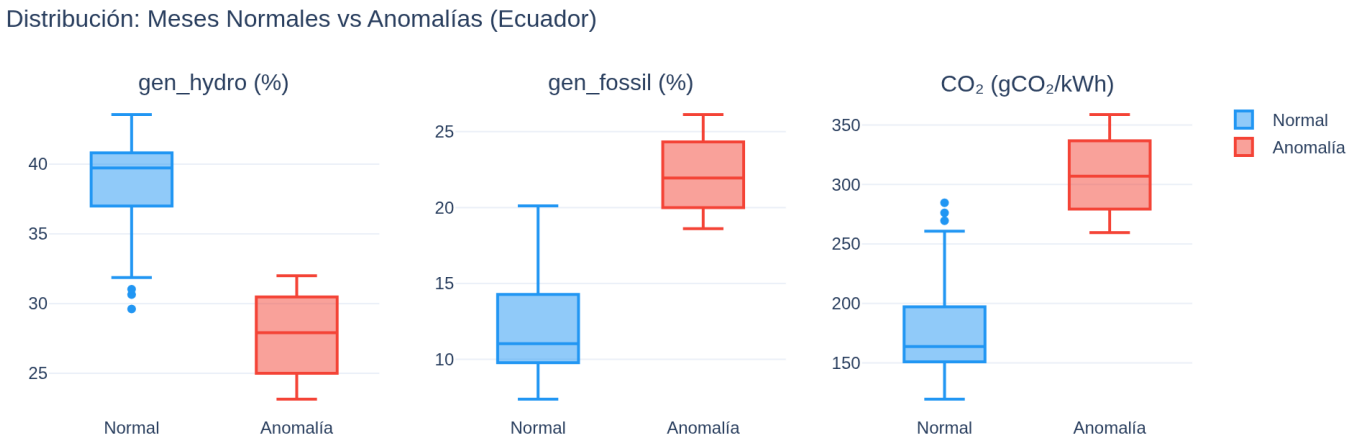
Heatmap sensibilidad



Scatter sensibilidad

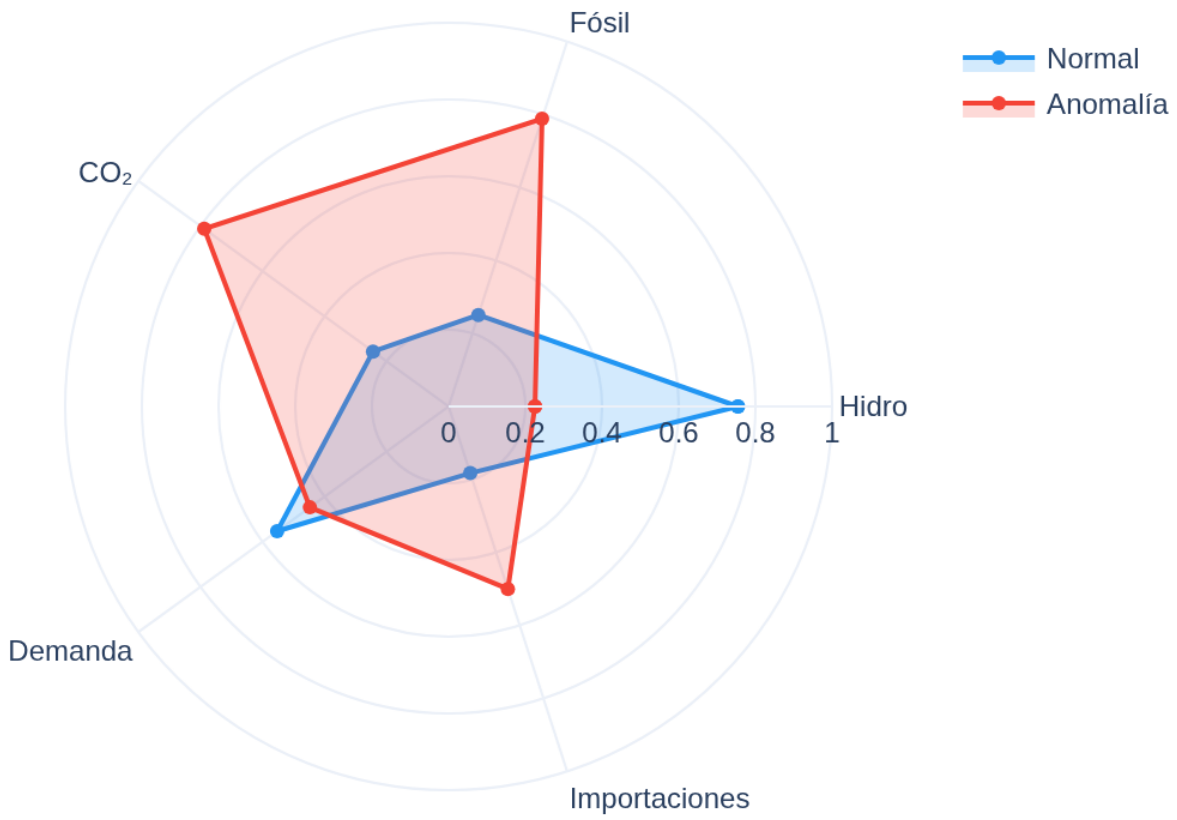
10. Validacion Estadistica

Mann-Whitney U: gen_hydro p=0.002**, gen_fossil p=0.003**, co2 p=0.008**
Cohen d > 1.2 (efecto GRANDE). Estabilidad: CV=0% en 20 semillas.



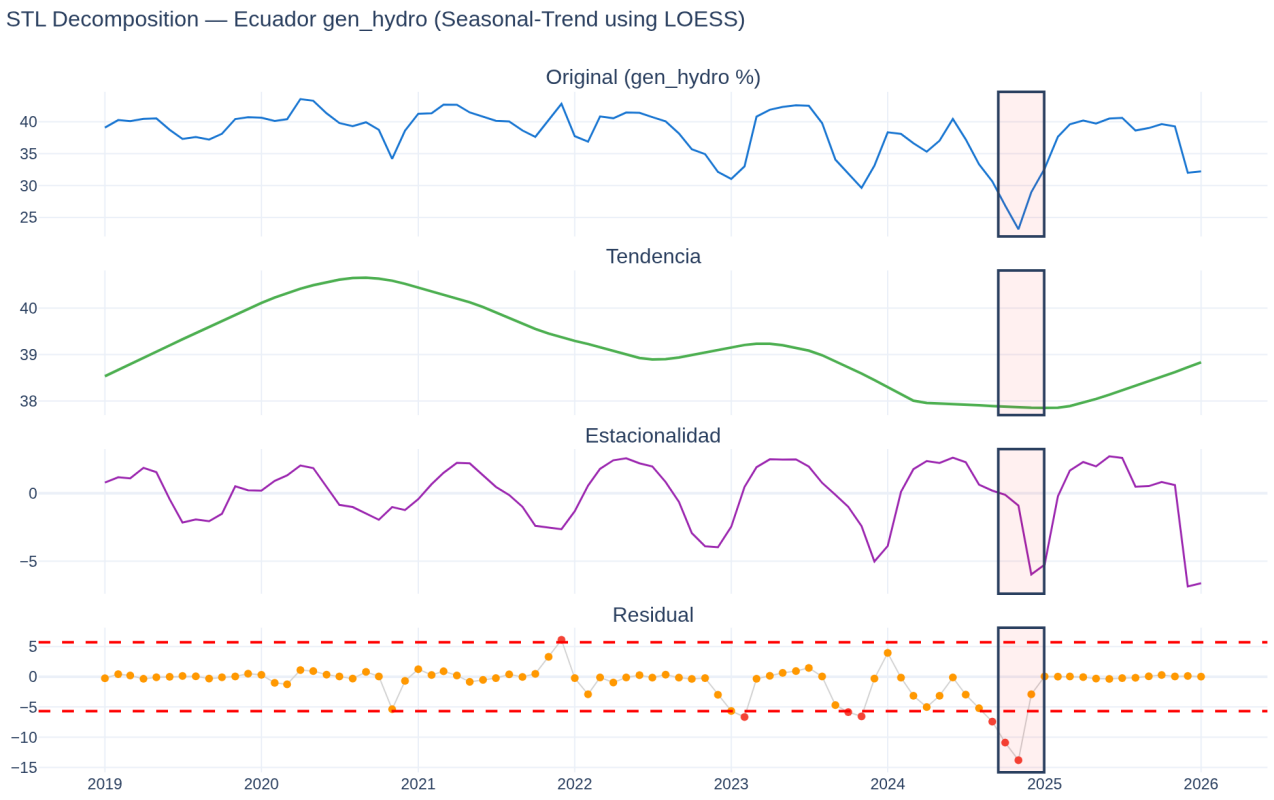
Normal vs anomalia

Perfil: Mes Normal vs Mes Anómalo (Ecuador)



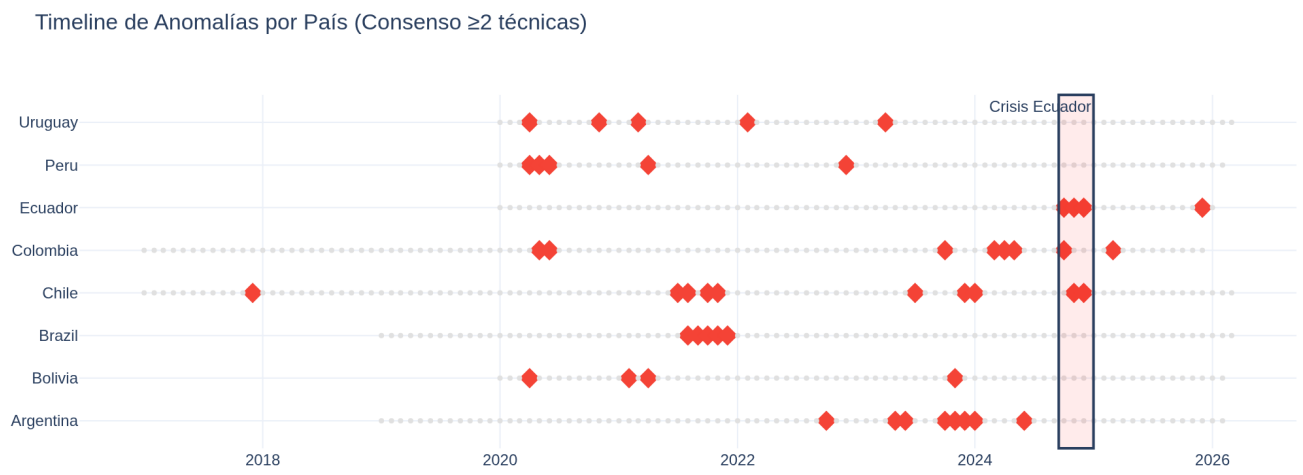
Perfil anomalia

11. STL Decomposition

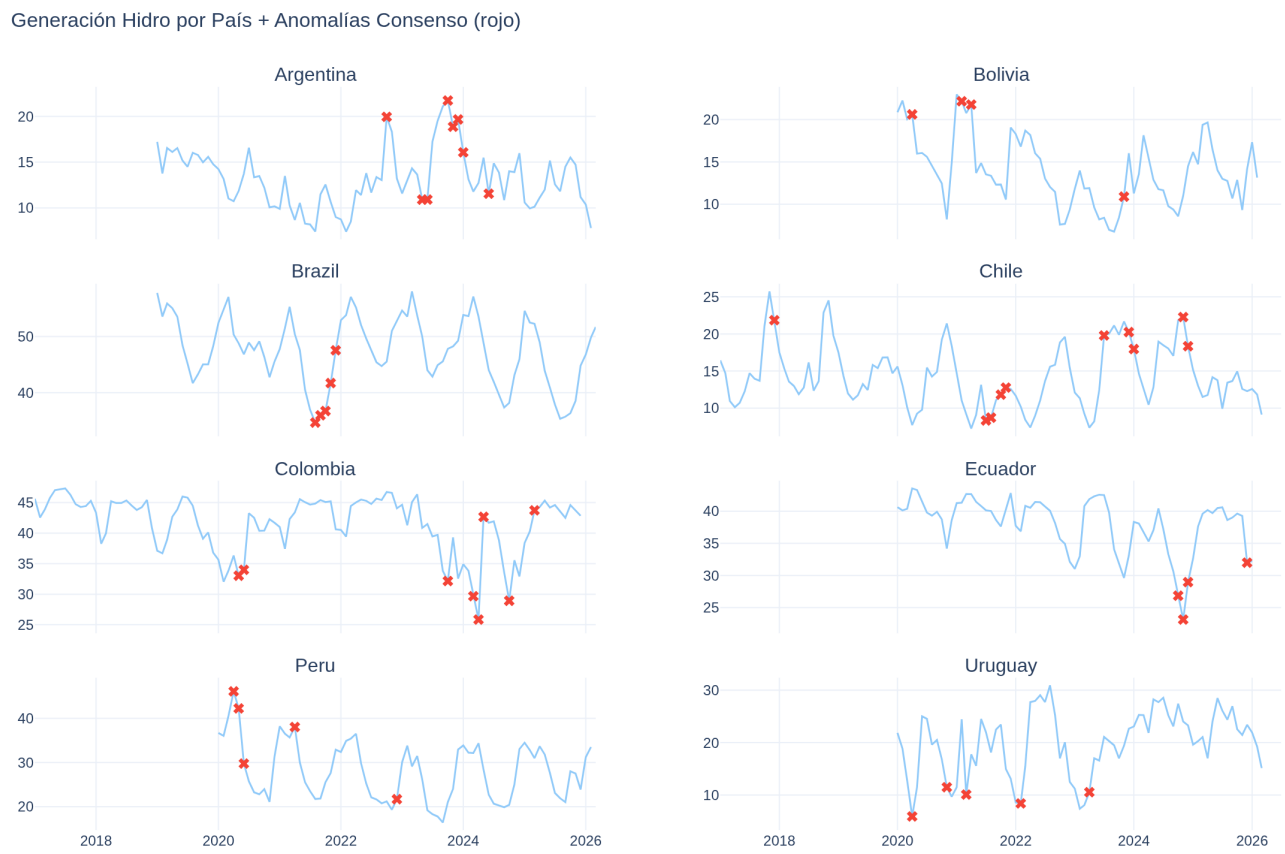


Descomposicion STL Ecuador

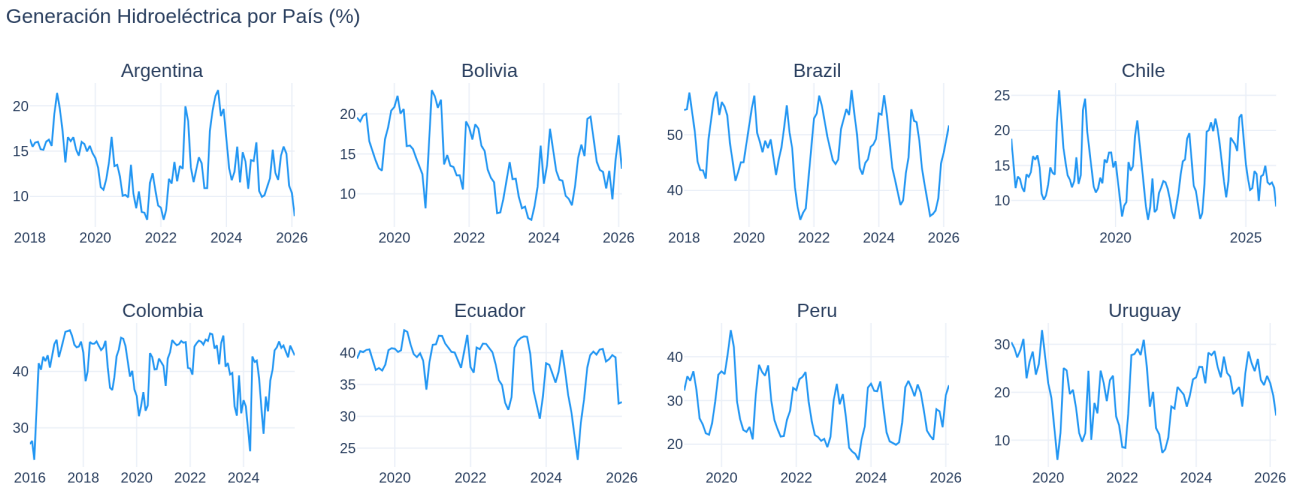
12. Analisis Multi-Pais



Timeline 8 paises



Hidro + anomalias



Comparativo hidro

13. Limitaciones

1. Datos mensuales (no diarios) suavizan crisis graduales
2. Ground truth parcial (crisis documentadas, no etiquetas expertas)
3. Faltan variables (embalses, precipitacion, precios spot)
4. 85 meses Ecuador es minimo viable
5. Consenso no mejora en paises con baja dependencia hidro (Chile, Argentina)
6. Bootstrap CI amplio por N pequeno [0.40, 1.00]

14. Aplicaciones

Sector energetico: CENACE, XM, COES (alerta temprana)
Clima: impacto El Nino en hidroelectricas
Academia: metodologia multi-tecnica, consenso
Politica: diversificacion energetica

Revistas potenciales: Energy Policy, Applied Energy, IEEE Access, Energies (MDPI)

15. Referencias

[1] Liu et al. (2008). Isolation Forest. IEEE ICDM.
[2] Cleveland et al. (1990). STL Decomposition.
[3] Page (1954). CUSUM. Biometrika.
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